School of Innovation and Management (SoIM)

Post Graduate Diploma in Management (PGDM)

PGDM (2025–27) | Trimester – 4

COURSE OUTLINE

Course Code:	Course Name: Forecasting and Demand Planning	Course Credits:03

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Course Coverage:

Forecasting and Demand Planning develops the ability to convert market and customer signals into useful forecasts, demand plans and operating decisions. The course integrates quantitative forecasting methods, Excel-based analysis, demand planning, managerial judgment and operations/supply-chain decision-making. The sequence follows: MARKET DEMAND → FORECAST → DEMAND PLAN → SUPPLY / OPERATIONS → DECISION → BUSINESS OUTCOME.

- Understand how demand uncertainty affects inventory, capacity, procurement, service and logistics decisions.
- Use quantitative forecasting as a baseline and combine it with Sales/Marketing intelligence and management assumptions.
- Connect forecasting and demand planning with operational choices under uncertainty and constraints.

Course Objectives

- Understand the role of forecasting and demand planning in operations and supply-chain decision-making.
- Develop competence in baseline, smoothing, trend/seasonality and causal forecasting using Excel.
- Evaluate forecast accuracy, bias and alternative models using evidence and business context.
- Translate statistical forecasts into consensus demand plans using commercial and market inputs.
- Convert demand plans into operational and supply-chain decisions involving inventory, capacity, procurement, service and logistics.
- Make and communicate evidence-based managerial decisions under uncertainty.

Course Learning Outcomes

CO	Course Learning Outcome
CO1	Explain the role of forecasting and demand planning in operations and supply-chain decision-making, and distinguish market demand, sales, forecast and demand plan.
CO2	Apply quantitative and Excel-based forecasting methods, evaluate forecast accuracy and bias, and develop a data-supported consensus demand plan using relevant business inputs.
CO3	Translate demand plans into operational and supply-chain decisions, evaluate trade-offs under demand uncertainty

	and operational constraints, and communicate an evidence-based managerial recommendation.

Program Outcomes

PO	Dimension	Description
PO1	Business Knowledge & Integration	Apply integrated business knowledge to understand demand, operations, supply-chain and managerial problems.
PO2	Analytical & Problem-Solving Skills	Use quantitative analysis, Excel and evidence to diagnose problems, evaluate alternatives and develop solutions.
PO3	Managerial Decision-Making & Communication	Make and communicate data-supported managerial decisions by evaluating trade-offs, uncertainty, operational implications and business value.

CO–PO Mapping

CO / PO	PO1	PO2	PO3
CO1	3	2	3
CO2	2	3	3
CO3	3	3	3

Assessment Tools for Learning Outcomes

Learning Outcome / Evidence	Assessment Criteria / Tools
CO1 – Conceptual and managerial understanding	Class participation, quizzes, caselet discussions and examinations.
CO2 – Quantitative forecasting capability	In-class Excel exercises, forecast accuracy/model-comparison assignment, quizzes and examination.
CO3 – Operational decision-making and communication	Demand-planning/supply-chain case, integrated simulation, executive forecast defense and final examination.

Pedagogy

The course will be covered across 12 sessions of 90 minutes each, totaling 18 hours. It will use interactive lectures, worked numerical examples, Excel-based exercises, managerial cases/caselets, forecast comparison, demand-planning scenarios and an integrated operations strategy simulation.

- Managerial cases and decision situations
- Worked numerical examples and Excel-based hands-on exercises
- Forecast comparison, accuracy and model selection
- Sales/Marketing consensus demand-planning scenarios
- Operational and supply-chain decision cases
- Integrated simulation and management recommendation

Session Plan – 12 Sessions × 90 Minutes

S. No.	Topic / Key Coverage	Pre-reading / Reference	Details of Pedagogy / Class Engagement / Caselets
1	Supply Chain Perspective: Why Demand Matters • Demand signal to operations • Lead time, inventory, capacity, service	Chopra & Meindl, Ch. 1 & 3	Interactive discussion on the end-to-end flow from customer/market signal to operational execution.
2	From Market / Sales Data to Forecasting • Market demand vs observed sales • Stock-outs/lost sales • Price, promotions, distribution and constraints	Chopra & Meindl, Ch. 7 Heizer, Render & Munson, Ch. 4	Data diagnosis exercise. Caselet: "The Empty Shelf Problem" – distinguish true demand from observed sales.
3	Baseline Forecasting Methods • Naïve and seasonal naïve • Simple and weighted moving averages	Chopra & Meindl, Ch. 7 Heizer, Render & Munson, Ch. 4	Worked numerical examples and Excel comparison of baseline methods.
4	Exponential Smoothing • Simple exponential smoothing • α and recursive updating • Responsiveness	Chopra & Meindl, Ch. 7 Heizer, Render & Munson, Ch. 4	Excel calculation; test alternative α values and discuss stability versus responsiveness.
5	Trend, Seasonality & Decomposition • Linear trend / least squares • Seasonal indices • Additive vs multiplicative	Chopra & Meindl, Ch. 7 Heizer, Render & Munson, Ch. 4	Dataset and Excel exercise to identify patterns and construct forecasts.
6	Forecast Accuracy, Bias & Model Selection • MAD, MSE, RMSE, MAPE, WAPE • CFE, bias, tracking signal	Chopra & Meindl, Ch. 7 Heizer, Render & Munson, Ch. 4	Calculation-reveal and model-selection discussion. Caselet: "Which Forecast Is Better?"
7	Demand Planning: From Statistical Forecast to Consensus Plan • Sales & Marketing inputs • Promotions, pricing, competition, new products	Chopra & Meindl, Ch. 7 & 9	Role-based demand review; build a consensus demand plan and document overrides/assumptions.
8	Causal Forecasting & Scenario Planning • Regression, coefficients and R^2 • Price, promotion, advertising • What-if scenarios	Chopra & Meindl, Ch. 7 Heizer, Render & Munson, Ch. 4	Excel regression and scenario exercise; quantify demand drivers and test assumptions.
9	From Demand Plan to Operational & Supply-Chain Decisions • Inventory, safety stock, production, procurement	Chopra & Meindl, Ch. 8, 9 & 12	Translate a consensus demand plan into an executable operating response.

	• Capacity, logistics, service and constraints		
10	Business Impact of Forecasting & Demand Planning • Inventory carrying cost, stock-outs, expediting • Working capital, capacity, revenue/contribution	Chopra & Meindl, Ch. 3 & 12 Faculty supplemental material	Quantitative business-impact exercise linking planning choices to inventory, service, capacity and working capital.
11	Operations Strategy: Decision-Making Under Demand Uncertainty • Service vs inventory vs cost • Capacity/flexibility, risk and constraints	Chopra & Meindl, Ch. 2, 8 & 12	Decision case comparing operating responses under alternative demand scenarios. Caselet: "The Flexible Factory Decision."
12	Integrated Demand Planning & Operations Strategy Simulation • Forecast selection, consensus plan, supply constraints and scenarios	Chopra & Meindl, Ch. 7, 8, 9 & 12	Integrated simulation: teams defend an end-to-end demand plan, operating response and managerial recommendation.

Recommended Text Book

Prescribed Text: Sunil Chopra and Peter Meindl, Supply Chain Management: Strategy, Planning, and Operation, 7th Edition, Pearson, 2019.

Primary prescribed reference for the course because it connects demand forecasting with supply-chain planning, strategic fit, aggregate planning, sales and operations planning, inventory and uncertainty. Chapter 7 is the principal forecasting reference; Chapters 8, 9 and 12 support the transition from forecast to demand plan and operational decisions.

Reference Text Books / Resources

- **Jay Heizer, Barry Render and Chuck Munson, Operations Management: Sustainability and Supply Chain Management, 12th Edition, Pearson India, 2017 – particularly Chapter 4, Forecasting**.
- B. Mahadevan, Operations Management, 3rd Edition, Pearson Education India – particularly Chapter 14, Demand Forecasting.
- Selected company/industry datasets, annual reports and faculty-developed caselets/exercises.
- Excel-based forecasting templates and instructor-provided quantitative datasets.

Course Evaluation

Evaluation is a continuous process at SoIM. Students will be given individual/group projects/ assignments in ----------. Keeping in line with continuous evaluation at SoIM, the evaluation scheme is as follows:

Areas	Weightage
(I) Continuous Evaluation: *(Quizzes: 10 Marks, Assignments, Group/Individual Projects: 30 Marks*	40%
(II) Mid Term Examination	20%
(III) Final Examination	40%
Total	**100 %**